CS 57800 – Statistical Machine Learning

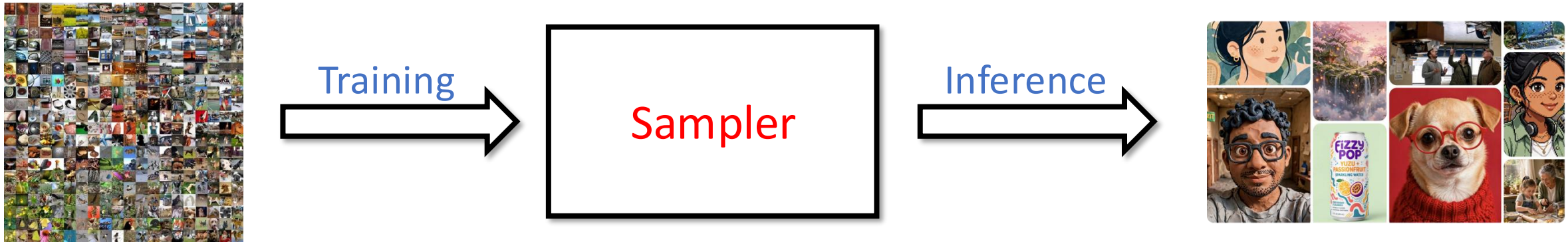
Lecture 4 Diffusion Model Theory

https://ruizhezhang.com/course_fall_2026.html

This slides is based on Sitan Chen's lecture notes.

Generative Modeling a.k.a. “Learning to Sample”

- **Input:** samples from data distribution (e.g., natural images)
- **Output:** a sampler that can generate samples from distributions that are close to the data distribution



Modern paradigm: progressive refinement

- Training: $x_0 \sim q$, “corrupt” x_0 to get x_t, corrupt more to get x_{t+h}, and train a neural network to predict x_t from x_{t+h}

Generative Modeling a.k.a. “Learning to Sample”

- **Input:** samples from data distribution (e.g., natural images)
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Training

Sampler

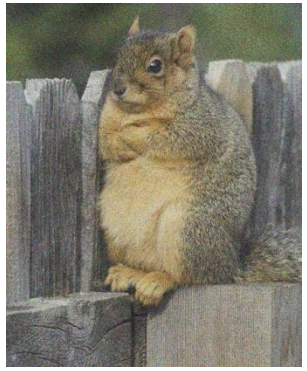
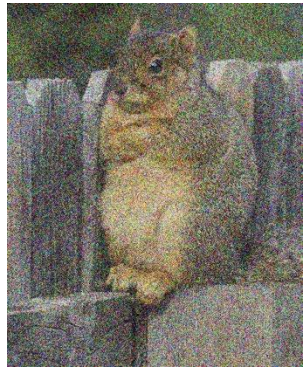
Inference

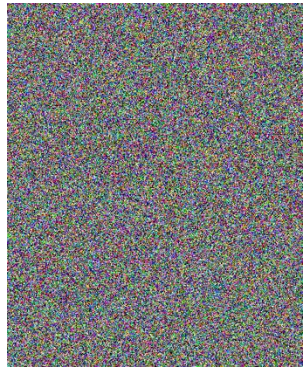


Modern paradigm: progressive refinement

- Inference: start from a “zero-information” point, progressively refine the point until it looks like a sample from the data distribution

Gaussian noising



- $x_0 \sim q$ (a sample from the data distribution)
- $x_t \sim \mathcal{N}(x_0, tI)$ a Gaussian noising
- As $t \to \infty$, x_t "forgets" the initial point x_0:

$$\mathrm{KL}(\mathcal{N}(x_0, tI) \| \mathcal{N}(0, tI)) = \frac{\|x_0\|^2}{2t}$$

- We can take a large T, sample $\tilde{x}_T \sim \mathcal{N}(0, TI)$ as x_T, and try to "recover" x_0 (?)

Gaussian denoising

- $x_0 \sim q$ (a sample from the data distribution)
- $x_t \sim \mathcal{N}(x_0, tI)$ a Gaussian noising
- Given x_t, what is the posterior over x_0?

$$\Pr[x_0|x_t] \propto \Pr[x_t|x_0] \cdot \Pr[x_0]$$

$$\propto \exp\left(-\frac{\|x_t - x_0\|^2}{2t}\right) \cdot q(x_0)$$

- When t is tiny, this posterior distribution is nearly a Gaussian concentrated around x_t
- Hard to characterize for a general data distribution q

Gaussian denoising

- $x_0 \sim q$ (a sample from the data distribution)
- $x_t \sim \mathcal{N}(x_0, tI)$ a Gaussian noising
- Given x_t, what is the **mean** of the posterior over x_0?
- **Tweedie's formula:** let q_t be the probability density of x_t, then

$$m_t(x_t) \coloneqq \mathbb{E}[x_0|x_t] = x_t + t\nabla \log q_t(x_t)$$

Bayesian correction

- Attempt for progressive refinement: $\mathbb{E}[x_{t-h}|x_t] = x_t + h\nabla \log q_t(x_t)$
  - Sample $\tilde{x}_T \sim \mathcal{N}(0, TI)$,
  - Compute $\tilde{x}_{T-h} \leftarrow \tilde{x}_T + h\nabla \log q_T(\tilde{x}_T)$
  - Compute $\tilde{x}_{T-2h} \leftarrow \tilde{x}_{T-h} + h\nabla \log q_{T-h}(\tilde{x}_{T-h})$
  - …
  - Output $\tilde{x}_0$

Gaussian denoising

- Recall that

$$\Pr[x_{t-h}|x_t] \propto \exp\left(-\frac{\|x_t - x_{t-h}\|^2}{2h}\right) \cdot q_{t-h}(x_{t-h})$$

- Most of the probability mass is around $O(\sqrt{h})$ away from x_t
- Tweedie's formula gives a correction $h\nabla \log q_t(x_t)$ of order h
- We can show that $\text{law}(x_{x-h}|x_t) \approx \mathcal{N}(x_t + h\nabla \log q_t(x_t), tI)$
  - Sample $\tilde{x}_T \sim \mathcal{N}(0, TI)$,
  - Sample $\tilde{x}_{T-h} \sim \mathcal{N}(\tilde{x}_T + h\nabla \log q_T(\tilde{x}_T), hI)$
  - Sample $\tilde{x}_{T-2h} \sim \mathcal{N}(\tilde{x}_{T-h} + h\nabla \log q_{T-h}(\tilde{x}_{T-h}), hI)$
  - …
  - Output $\tilde{x}_0$

Diffusion sampling

- Sample $\tilde{x}_T \sim \mathcal{N}(0, TI)$,
- Sample $\tilde{x}_{T-h} \sim \mathcal{N}(\tilde{x}_T + h\nabla \log q_T(\tilde{x}_T), hI)$
- Sample $\tilde{x}_{T-2h} \sim \mathcal{N}(\tilde{x}_{T-h} + h\nabla \log q_{T-h}(\tilde{x}_{T-h}), hI)$
- …
- Output $\tilde{x}_0$

How to get $\nabla \log q_t$, the **score** of q_t?

$$m_t(x_t) \coloneqq \mathbb{E}[x_0|x_t] = x_t + t\nabla \log q_t(x_t)$$

- Score estimation: train a neural network on (x_0, t, x_t):

$$m_t(\cdot) \coloneqq \arg\min_f \|x_0 - f(x_t)\|^2$$

Diffusion sampling

An alternative approach

- Sample $\tilde{x}_T \sim \mathcal{N}(0, TI)$,
- Compute $\tilde{x}_{T-h} = \tilde{x}_T + \frac{1}{2} h \nabla \log q_T(\tilde{x}_T)$
- Compute $\tilde{x}_{T-2h} = \tilde{x}_{T-h} + \frac{1}{2} h \nabla \log q_{T-h}(\tilde{x}_{T-h})$
- …
- Output $\tilde{x}_0$

Diffusion sampling

We made several assumptions/heuristics:

- $\tilde{x}_T \sim \mathcal{N}(0, TI)$ is pretended to be x_T
- $\text{law}(x_{x-h}|x_t) \approx \mathcal{N}(x_t + h\nabla \log q_t(x_t), tI)$
- h is not set to be infinitesimally small
- Score estimation with finite data / compute

Stochastic calculus is the mathematical language to study generative modeling in a continuous space and continuous time

Brownian motion

$(\boldsymbol{B}_t)_{t\geq 0} \subset \mathbb{R}^d$ is a stochastic process (random sequence of vectors indexed by t) such that:

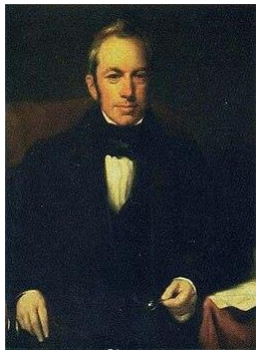
- Starts at origin: $\boldsymbol{B}_0 = 0$ is the origin in $\mathbb{R}^d$
- Continuous paths: With probability 1, $t \mapsto \boldsymbol{B}_t$ is continuous
- Independent increments: For all $0 = t_0 < t_1 < \cdots < t_k$, the random vectors $\boldsymbol{B}_{t_{i+1}} - \boldsymbol{B}_{t_i}$ for $0 \leq i < k$ are mutually independent
- Gaussian increments: For all $0 \leq s \leq t$, $\boldsymbol{B}_t - \boldsymbol{B}_s \sim N(0, (t-s)I)$

The probability space that Brownian motion is defined on is denoted $\{\mathcal{F}_t\}_{t\geq 0}$, a filtration satisfying $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $0 \leq s \leq t$. We say that Brownian motion is a stochastic process adapted to the filtration $\{\mathcal{F}_t\}_{t\geq 0}$

$\mathcal{F}_t$ contains the information of the randomness used up to time t

Brownian motion

1827 — 1900 — 1905 — 1908 — 1923



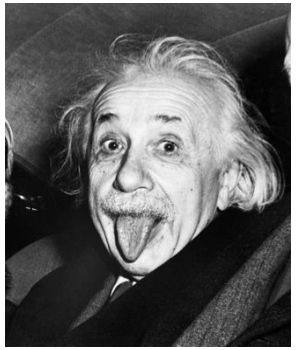
Robert Brown

Observes erratic motion of pollen grains in water under microscope



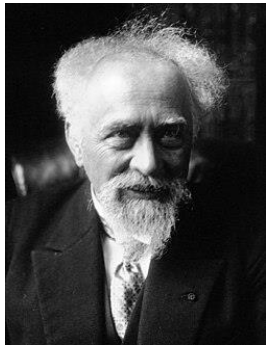
Louis Bachelier

First mathematical treatment of Brownian motion, used to model stock prices



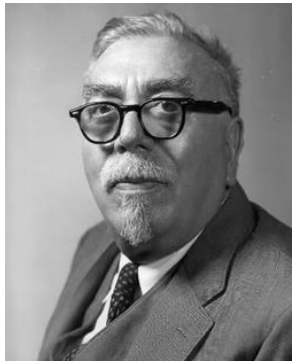
Albert Einstein

Explains *physical* mechanism behind Brownian motion, used to estimate Avogadro's number



Jean Perrin

Experimentally confirms Einstein's predictions, receives Nobel Prize

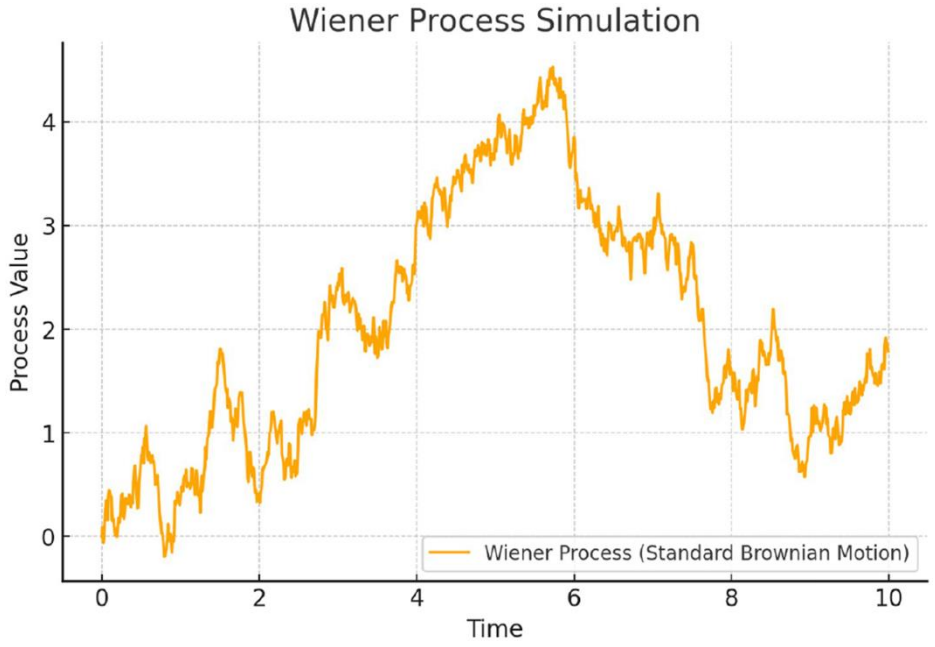


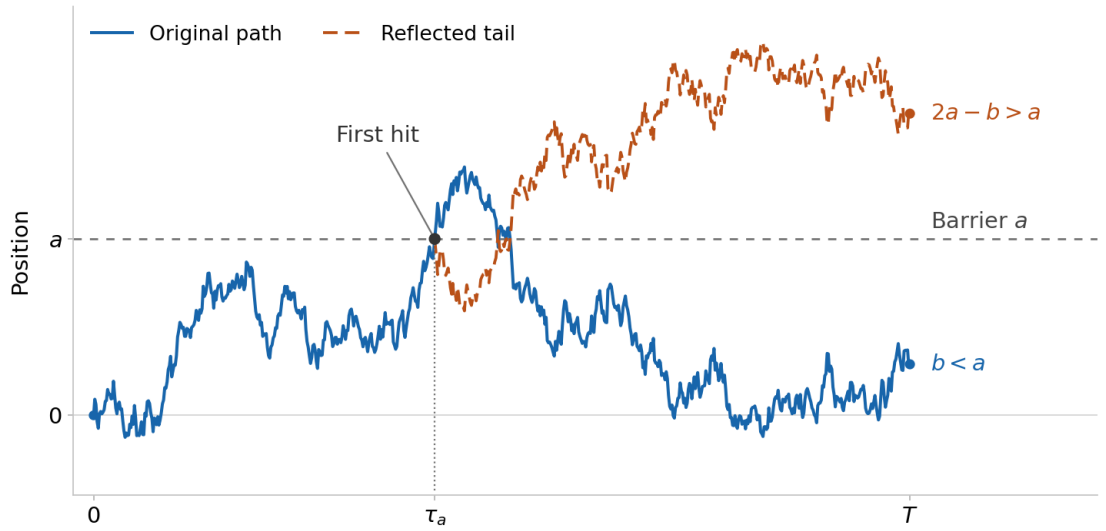
Norbert Wiener

Develops fully rigorous mathematical theory for Brownian motion

Brownian motion

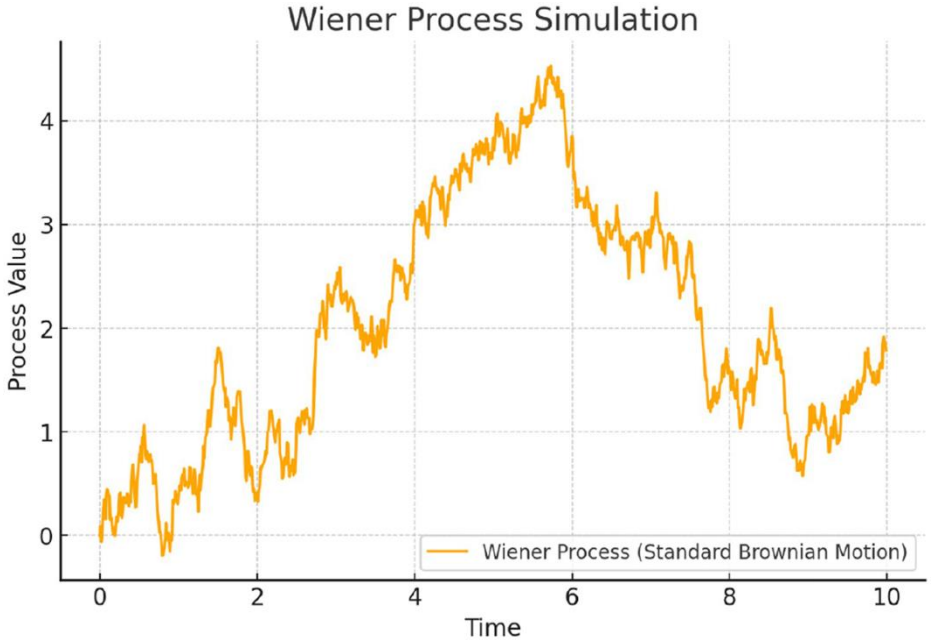
- Over a time window t, $\boldsymbol{B}_t$ moves $O\left(\sqrt{td}\right)$
- Although is $t \mapsto \boldsymbol{B}_t$ is a.s. continuous, it's a.s. nowhere differentiable
- Induces a probability measure over "path space" $[0,T]\times\mathbb{R}^d$ (Wiener measure)
- The Wiener measure is unique, but pathwise realization is **not** unique
  - Suppose $\boldsymbol{B}_t$ is some realization
  - $-\boldsymbol{B}_t$ is also a Brownian motion
  - Reflection principle: At the first time $\boldsymbol{B}_t$ hits a hyperplane H, reflect $\boldsymbol{B}_t$ about H

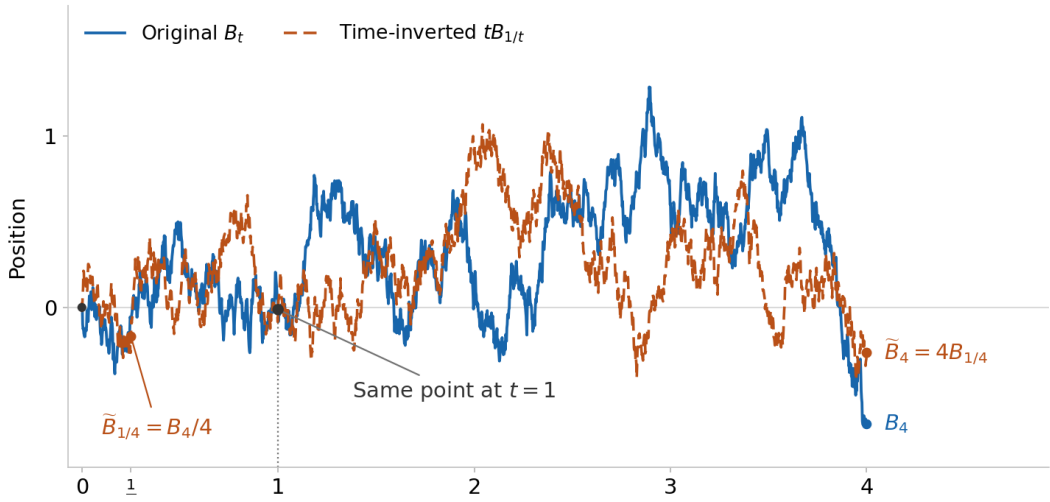




Brownian motion

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  - Reflection principle: At the first time $\boldsymbol{B}_t$ hits a hyperplane H, reflect $\boldsymbol{B}_t$ about H
  - Time inversion: $t\boldsymbol{B}_{1/t}$





Stochastic integrals

Given continuous process $(\mathbf{H}_t)_{t\geq 0} \in \mathbb{R}^{d\times d}$ adapted to the filtration generated by $(\boldsymbol{B}_t)_{t\geq 0}$, the Itô integral

$$\mathbf{x}_t = \int_0^t \mathbf{H}_s \, \mathrm{d}\boldsymbol{B}_s$$

is the stochastic process whose value at time t is the random vector given by taking the probability limit over meshes $P = \{t_k\} \subset [0, t]$ such that $0 = t_0 < \cdots < t_{|P|} = t$:

$$\int_0^t \mathbf{H}_s \, \mathrm{d}\mathbf{B}_s = \lim_{\|P\|_{\text{gap}}\to 0} \sum_{j=1}^{|P|} \mathbf{H}_{t_j} \cdot \left(\boldsymbol{B}_{t_{j+1}} - \boldsymbol{B}_{t_j}\right)$$

where $\|P\|_{\text{gap}} \coloneqq \max_{j\in[|P|]} \left|t_j - t_{j-1}\right|$

Stochastic integrals

The Itô integral $\mathbf{x}_t = \int_0^t \mathbf{H}_s \, \mathrm{d}\boldsymbol{B}_s$ is a continuous martingale

- Continuity: With prob. 1 over the randomness, $t \mapsto \mathbf{x}_t$ is continuous
- Martingale: $\mathbb{E}[\mathbf{x}_t | \mathcal{F}_s] = \mathbf{x}_s \iff \mathbb{E}\left[\int_s^t \mathbf{H}_s \, \mathrm{d}\boldsymbol{B}_s \,\middle|\, \mathcal{F}_s\right] = 0$
- Itô isometry:

$$\mathbb{E}\left[\left\|\int_0^t \mathbf{H}_s \, \mathrm{d}\boldsymbol{B}_s\right\|^2\right] = \mathbb{E}\left[\int_0^t \|\mathbf{H}_s\|_F^2 \, \mathrm{d}s\right]$$

$\mathbb{E}[(a_1 g_1 + a_2 g_2 + \cdots + a_k g_k)^2] = a_1^2 + \cdots + a_k^2$ for $g_1, \ldots, g_k \sim N(0,1)$

"$\mathrm{d}\boldsymbol{B}_t^2 = \mathrm{d}t$" (because $\boldsymbol{B}_{t+\mathrm{d}t} - \boldsymbol{B}_t \sim \mathcal{N}(0, \mathrm{d}t)$)

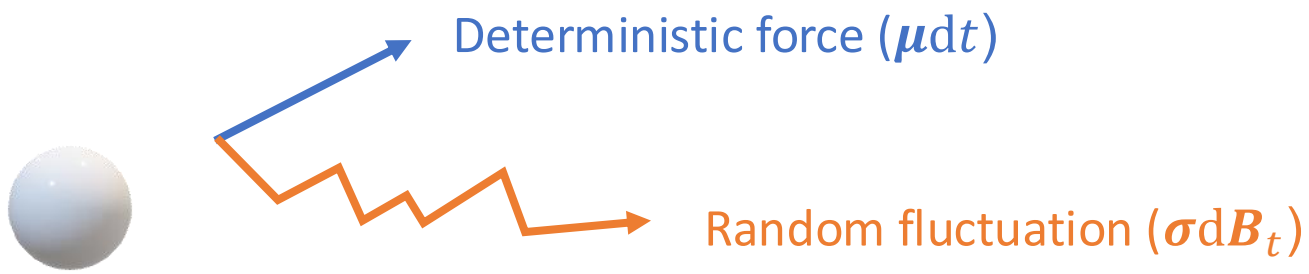
Drift-diffusion processes

A drift-diffusion process $\{\mathbf{x}_t\}_{t\geq 0}$ on $\mathbb{R}^d$ is driven by a time-dependent drift $\boldsymbol{\mu}_t\colon \mathbb{R}^d \to \mathbb{R}^d$ and a diffusion coefficient $\boldsymbol{\sigma}_t\colon \mathbb{R}^d \to \mathbb{R}^{d\times d}$, and captured by the stochastic differential equation (SDE):

$$\mathrm{d}\mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_t)\mathrm{d}t + \boldsymbol{\sigma}_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$$

We can write this stochastic process in integral form (Itô process):

$$\mathbf{x}_t = \mathbf{x}_0 + \int_0^t \boldsymbol{\mu}_t(\mathbf{x}_t)\mathrm{d}t + \int_0^t \boldsymbol{\sigma}_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$$



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- Technically need to check such a process exists and is uniquely defined

Theorem. If $\boldsymbol{\mu}_t$ and $\boldsymbol{\sigma}_t$ have a finite Lipschitz constant uniformly over $t \in [0, T]$, then the SDE solution exists and is unique

Drift-diffusion processes

Ornstein-Uhlenbeck process

$$\mathrm{d}\mathbf{x}_t = -\mathbf{x}_t \mathrm{d}t + \sqrt{2}\mathrm{d}\boldsymbol{B}_t$$

- Without the diffusion term, the ODE $\mathrm{d}x_t = -x_t \mathrm{d}t$ has solution $x_t = e^{-t}x_0$
- We can solve this SDE in a similar way as solving the ODE:

$$\mathrm{d}(e^t \mathbf{x}_t) = e^t \mathbf{x}_t \mathrm{d}t + e^t \mathrm{d}\mathbf{x}_t = e^t \mathbf{x}_t \mathrm{d}t + e^t\left(-\mathbf{x}_t \mathrm{d}t + \sqrt{2}\mathrm{d}\boldsymbol{B}_t\right) = \sqrt{2}e^t d\boldsymbol{B}_t$$

$$\mathbf{x}_t = e^{-t}\mathbf{x}_0 + \underbrace{\int_0^t \sqrt{2}e^{-(t-r)}\mathrm{d}\boldsymbol{B}_r}$$

Distributed as $\mathcal{N}(0, \beta_t^2 I)$ with $\beta_t^2 = 1 - e^{-2t}$

- Solution: $\mathbf{x}_t \sim \mathcal{N}(e^{-t}\mathbf{x}_0, (1 - e^{-2t})I)$
- As $t \to \infty$, $\mathrm{Law}(\mathbf{x}_t) \to \mathcal{N}(0, I)$, i.e. the standard Gaussian is the stationary distribution

Drift-diffusion processes

(Overdamped) Langevin diffusion

For a potential / Hamiltonian $V\colon \mathbb{R}^d \to \mathbb{R}$, define

$$\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\mathrm{d}t + \sqrt{2}\mathrm{d}\boldsymbol{B}_t$$

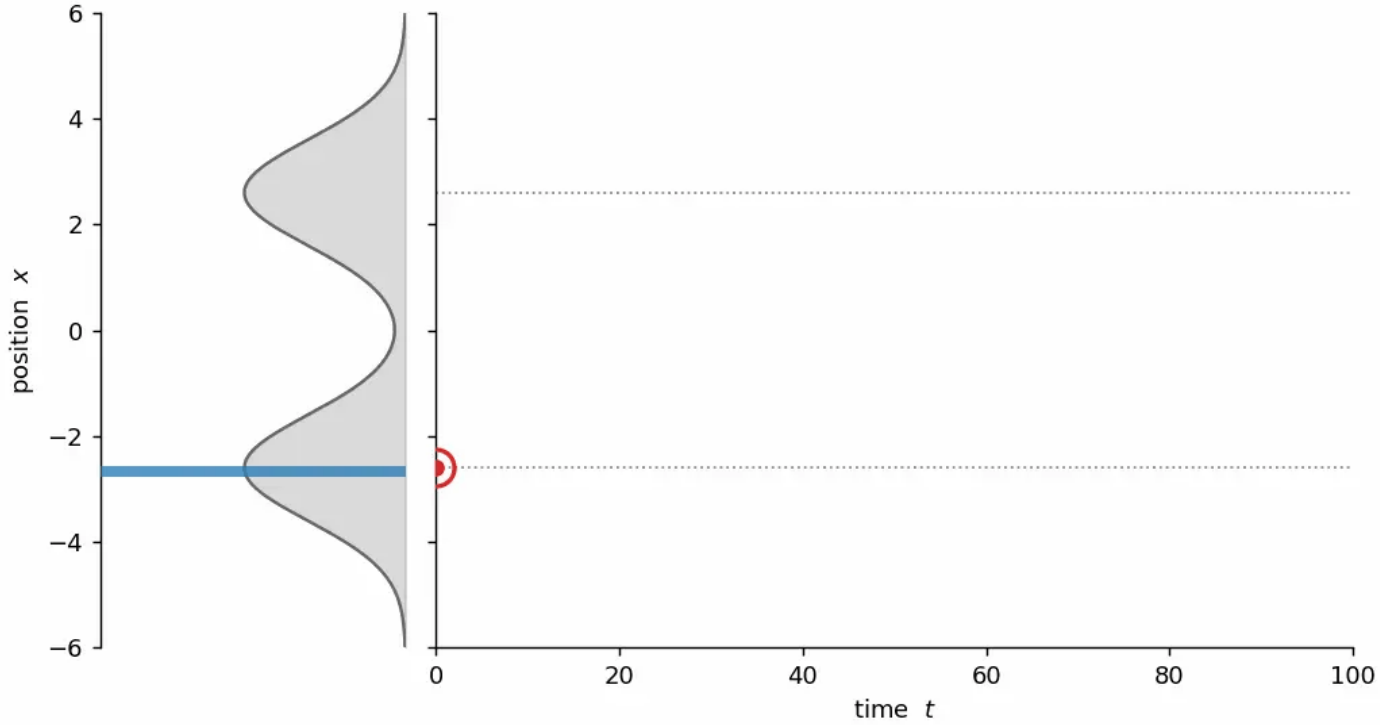
- "Gradient descent with noise"
- Stationary distribution: as $t \to \infty$, $\mathrm{Law}(\mathbf{x}_t) \to \pi(x) = \frac{1}{Z}e^{-V(x)}$, i.e. the Gibbs distribution
- The drift $-\nabla V$ is the score function of the target π
- OU is a special case with $V(x) = \frac{1}{2}\|x\|^2$

Sampling from Gibbs Distributions

$$\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\mathrm{d}t + \sqrt{2}\mathrm{d}\boldsymbol{B}_t$$

Euler–Maruyama discretization:

$$x_{t+h} = x_t - h\nabla V(x_t) + \sqrt{2h}\gamma, \qquad \gamma \sim \mathcal{N}(0, I)$$

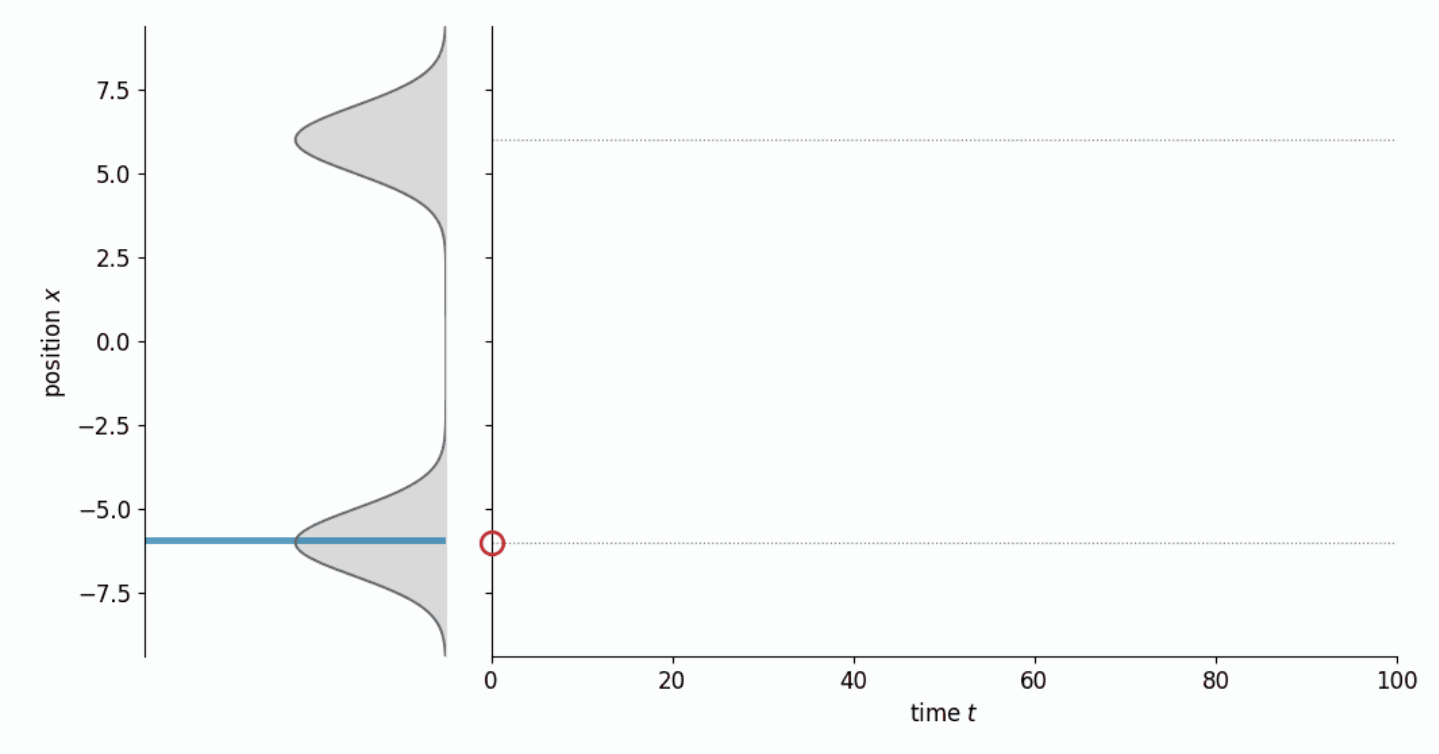


Sampling from Gibbs Distributions

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Euler–Maruyama discretization:

$$x_{t+h} = x_t - h\nabla V(x_t) + \sqrt{2h}\gamma, \qquad \gamma \sim \mathcal{N}(0, I)$$



Stochastic chain rule

$$\mathrm{d}f(g) = f'(g) \cdot \mathrm{d}g$$

- For an ODE $\mathrm{d}x_t = b_t(x_t)\mathrm{d}t$, for any differentiable f, we have

$$\mathrm{d}f(x_t) = \langle \nabla f(x_t), b_t(x_t) \rangle \mathrm{d}t$$

Itô's lemma. Let $f: \mathbb{R}^d \to \mathbb{R}$ be twice-differentiable, and suppose $\{\mathbf{x}_t\}_{t \geq 0}$ follows the SDE:

$$\mathrm{d}\mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_t)\mathrm{d}t + \boldsymbol{\sigma}_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$$

Then $\{f(\mathbf{x}_t)\}_{t \geq 0}$ is a stochastic process following the SDE:

$$\mathrm{d}f(\mathbf{x}_t) = \left(\langle \nabla f(\mathbf{x}_t), \boldsymbol{\mu}_t(\mathbf{x}_t) \rangle + \frac{1}{2} \langle \nabla^2 f(\mathbf{x}_t), \boldsymbol{\sigma}_t(\mathbf{x}_t) \boldsymbol{\sigma}_t(\mathbf{x}_t)^\top \rangle \right) \mathrm{d}t + \langle \nabla f(\mathbf{x}_t), \boldsymbol{\sigma}_t(\mathbf{x}_t) \mathrm{d}\boldsymbol{B}_t \rangle$$

"$\mathrm{d}\boldsymbol{B}_t^2 = \mathrm{d}t$"

Stochastic chain rule

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Itô's lemma. Let $f: \mathbb{R}^d \to \mathbb{R}$ be twice-differentiable, and suppose $\{\mathbf{x}_t\}_{t \geq 0}$ follows the SDE:

$$\mathrm{d}\mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_t)\mathrm{d}t + \boldsymbol{\sigma}_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$$

Then $\{f(\mathbf{x}_t)\}_{t \geq 0}$ is a stochastic process following the SDE:

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- Consider the Langevin diffusion: $\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\, \mathrm{d}t + \sqrt{2}\, \mathrm{d}\boldsymbol{B}_t$

$$\mathrm{d}f(\mathbf{x}_t) = (-\langle \nabla f, \nabla V \rangle + \Delta f)\mathrm{d}t + \sqrt{2} \langle \nabla f, \mathrm{d}\boldsymbol{B}_t \rangle, \qquad \Delta f \coloneqq \mathrm{tr}[\nabla^2 f]$$